Pharmaceutical Sales Data Analysis: From Raw Data to Actionable Business Insights

Graduation Project Report

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Introduction

This report details the methodology, analysis, and strategic findings of a comprehensive data science project focused on pharmaceutical sales performance. In the complex and highly regulated pharmaceutical distribution sector, understanding sales performance across varied regions and customer types is critical for strategic planning. This project focuses on analyzing an extensive sales dataset to provide a granular view of drug product performance. The aim is to move beyond mere transaction tracking to reveal core drivers of profitability and market penetration, informing stakeholders on optimal allocation of resources and channel focus.

The project successfully established a robust, multi-tool data pipeline, beginning with rigorous data cleaning using SQL, followed by exploratory data analysis and visualization in Excel, and culminating in the development of interactive business intelligence dashboards using Power BI. The subsequent sections of this report detail the execution of this pipeline and present the actionable insights derived from the analysis.

Acknowledgements

The successful completion of this graduation project would not have been possible without the guidance and support of several key individuals and institutions.

We extend our deepest gratitude to the **DEPI Team** for providing the foundational framework, resources, and continuous support necessary for this project. Their commitment to fostering data science excellence was instrumental in our success.

A special and heartfelt thank you is reserved for our helpful and excellent instructor, **Menna Selim**. Her continuous mentorship, expert guidance, and insightful feedback throughout the project lifecycle were invaluable. Her dedication ensured the rigor and quality of our analysis and final deliverables.

Finally, we thank our families and peers for their unwavering encouragement and support.

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1. Executive Summary

The analysis highlights key patterns in regional sales, customer behavior, and product performance. Initial findings show noticeable differences across regions, strong contributions from hospital and specialty pharmacy channels, and clear growth opportunities in specific markets.

Overall, the project provides a data-driven view of sales trends and offers practical recommendations to enhance performance, strengthen high-value channels, and support future business planning.

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2. Project Overview

2.1 Background

This section highlights the need for deep insight into pharmaceutical sales performance and distribution channels.

- **Customized Content:** "In the complex and highly regulated pharmaceutical distribution sector, understanding sales performance across varied regions and customer types is critical for strategic planning. This project focuses on analyzing an extensive sales dataset to provide a granular view of drug product performance. The aim is to move beyond mere transaction tracking to reveal core drivers of profitability and market penetration, informing stakeholders on optimal allocation of resources and channel focus."

2.2 Objectives

These specific, measurable objectives address the multi-faceted nature of your analysis (Companies, Regions, Channels, Sales Team).

- **Customized Content:**I. To clean and structure the raw sales data to create a reliable foundation for all subsequent analyses.
  II. To develop a comprehensive set of dashboards targeting distinct business areas (e.g., Company Performance, Regional Sales Dynamics, Channel Effectiveness, and Sales Representative Metrics).
  III. To identify and compare the sales performance and profitability drivers across major global regions (North America, Europe, Asia, Oceania, South America).
  IV. To understand customer behavior by analyzing the contribution of different distribution channels (Hospital, Retail Pharmacy, Government, etc.) to overall revenue.
  V. To provide data-driven recommendations aimed at optimizing sales strategies and improving monthly sales performance tracking.

2.3 Scope

This defines the boundaries of the analysis, specifically mentioning the dataset and the reporting outcome.

- **Dataset Overview:** The project utilizes a dataset comprising ~3,000 rows and 23 columns of pharmaceutical product sales data.
- **Customized Content:**
  - **Included:** Analysis is restricted to the provided raw sales data, encompassing product categories, sales volumes, channels, and geographical distribution (as listed: North America, Europe, Asia, Oceania, South America). The primary deliverable is a suite of interactive analytical dashboards.
  - **Excluded:** The scope does not cover forecasting beyond the existing data period, nor does it include primary data collection (e.g., market research) or advanced predictive modeling using machine learning algorithms.

2.4 Tools Used

Tool / Category	Technology	Primary Purpose
Data Extraction & Structuring	SQL	Initial data querying, extraction, data cleansing, transformation, and creating structured tables (ETL process).
Data Transformation & Cleaning	Excel	Handling initial data profiling, cleaning, and preliminary data manipulation (where SQL or Power BI was less efficient).
Visualization & Reporting	Power BI	Developing the interactive, comprehensive dashboards for various stakeholders and the final presentation.
Advanced Analysis	Tableau	Used for specific exploratory data analysis (EDA) and cross-validation of visualizations.

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3. Data Source & Description

3.1 Data Origin

The data for this analysis was compiled from multiple sources reflective of real-world pharmaceutical distribution operations. The dataset primarily originates from Internal Distribution Company Records and was extracted from various formats, including Raw files and consolidated Excel sheets.

Data Handling: The data was initially complex, requiring significant use of SQL for querying, cleaning, and standardization to ensure consistency across all regional and channel entries before loading into the visualization tool.

3.2 Dataset Structure

The final analytical dataset provides a comprehensive snapshot of sales activities, enabling granular analysis.

Metric	Details
Total Records (Rows)	Approximately 3,000
Total Features (Columns)	23
Time Period	[The last 12 months (Jan 2024 - Dec 2024)]
Geographic Coverage	Global, covering five main regions: North America, Europe, Asia, Oceania, and South America.
Data Type	Pharmaceutical Product Sales Data (Transactional and Categorical).

3.3 Key Variables

The analysis focuses on several critical dimensions and measures that define sales performance and customer segmentation.

Variable Category	Examples (Dimensions & Measures)	Purpose in Analysis
Geographical	Region, Country, Area Code	Analyzing sales distribution, identifying top-performing markets, and understanding regional market differences.
Distribution Channel	Channel, Channel_Description (Hospital, Retail Pharmacy, Government, Online, etc.)	Evaluating the efficiency and sales contribution of each route to market.
Product/Item	Product Name, Drug Type, Manufacturer	Assessing the performance of specific product categories and competitor analysis.
Sales Performance	Sales Volume, Revenue, Profit Margin, Price per Unit	Core metrics used to measure financial performance and profitability drivers.
Sales Force	Sales Person ID, Sales Team	Tracking individual and team performance metrics.

4. Data Cleaning & Preparation

This phase ensured the quality, consistency, and suitability of the raw data for analysis and visualization. The process involved systematic steps to handle imperfections and optimize the dataset structure.

4.1 Data Inspection

This report summarizes the inspection of the dataset to assess its quality, structure, and readiness for analysis. Key issues, patterns, and recommendations are highlighted to support accurate decision-making.

The code:

```
--Data Inspection----------------

SELECT *
FROM [FACT_SALES]
WHERE
Quantity IS NULL
OR Unit_Price IS NULL
OR Customer_ID IS NULL
OR Order_Priority IS NULL
OR City_ID IS Null
OR LOWER(LTRIM(RTRIM(Order_Priority))) NOT IN ('high','medium','low')
OR (
TRY_CONVERT(date, Transaction_Date, 23) IS NULL
AND TRY_CONVERT(date, Transaction_Date, 105) IS NULL
)
OR Quantity > (
SELECT AVG(Quantity) + 3 * STDEV(Quantity) FROM Fact_Sales

);
```

4.2 Data Calculation

```
-----Counting Total rows containing Errors in Data----

SELECT
COUNT(*) AS Total_Error_Rows
FROM Fact_Sales
WHERE
```

```
Quantity IS NULL
OR Unit_Price IS NULL
OR Customer_ID IS NULL
OR Order_Priority IS NULL
OR LOWER(LTRIM(RTRIM(Order_Priority))) NOT IN ('high','medium','low')
OR (
TRY_CONVERT(date, Transaction_Date, 23) IS NULL
AND TRY_CONVERT(date, Transaction_Date, 105) IS NULL
)
OR Quantity > (
SELECT AVG(Quantity) + 3 * STDEV(Quantity) FROM Fact_Sales WHERE Quantity
IS NOT NULL
);
-------------------------------------------------------------------------
---------
-----Counting Total rows containing missing Data----
SELECT
COUNT(*) AS Total_Rows,
SUM(CASE WHEN Quantity IS NULL THEN 1 ELSE 0 END) AS Missing_Quantity,
SUM(CASE WHEN Unit_Price IS NULL THEN 1 ELSE 0 END) AS Missing_UnitPrice,
SUM(CASE WHEN Customer_ID IS NULL THEN 1 ELSE 0 END) AS Missing_Customer,
SUM(CASE WHEN Order_Priority IS NULL THEN 1 ELSE 0 END) AS
Missing_Priority,
SUM(CASE WHEN City_ID IS NULL THEN 1 ELSE 0 END) AS Missing_City
```

```
FROM Fact_Sales;
```

4.3 Missing Values

```
---- Correction of Null values in Order priority
```

```
SELECT [Transaction_ID],[Customer_ID],[City_ID],[Order_Priority]
FROM Fact_Sales
WHERE Order_Priority IS NULL;
UPDATE Fact_Sales
SET Order_Priority = 'Medium'
WHERE Order_Priority IS NULL;
-----------------------------------------------------
------------ Correction of Null values in Customer_ID
SELECT [Transaction_ID],[City_ID],[Order_Priority],[Customer_ID]
FROM Fact_Sales
WHERE Customer_ID IS NULL;
UPDATE Fact_Sales
SET Customer_ID = 0
WHERE Customer_ID IS NULL;
---------------------------------------------------
----------------Correction of Null values in uint_Price
```

```
SELECT
[Transaction_ID],[City_ID],[Order_Priority],[Customer_ID],[Unit_Price]
FROM Fact_Sales
WHERE Unit_Price IS NULL;
UPDATE Fact_Sales
SET Unit_Price = (SELECT AVG(Unit_Price) FROM Fact_Sales)
WHERE Unit_Price IS NULL;
------------------------------------------------------------------------
-
----------------Correction of Null values in uint_quantity
SELECT
[Transaction_ID],[City_ID],[Order_Priority],[Customer_ID],[Unit_Price],[T
otal_Sales],
[Quantity]
FROM Fact_Sales
WHERE Quantity IS NULL;
UPDATE Fact_Sales
SET Quantity = Total_Sales/ Unit_Price
WHERE Quantity IS NULL;
```

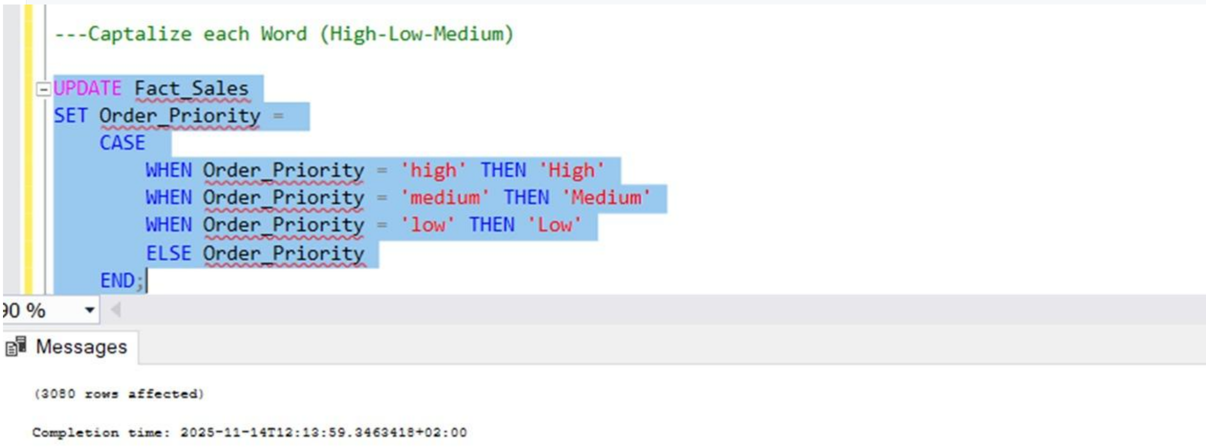
```
------------------------------------------------------------------------
-
```

4.4 Standardization & Transformation

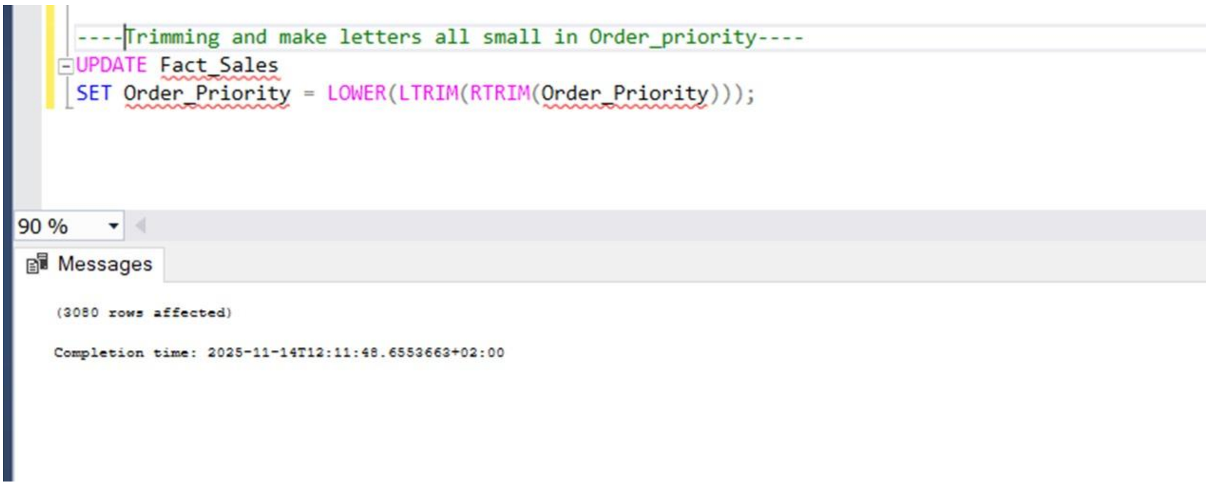
```
-------Correction of Non-Customized Data in Order_priority
```

```
SELECT TOP 50
Transaction_ID,
Customer_ID,
Order_Priority,
Total_Sales
FROM Fact_Sales;
-------------------------------------------------------------
----Trimming and make letters all small in Order_priority----
UPDATE Fact_Sales
SET Order_Priority = LOWER(LTRIM(RTRIM(Order_Priority)));
---Captalize each Word (High-Low-Medium)
UPDATE Fact_Sales
SET Order_Priority =
CASE
WHEN Order_Priority = 'high' THEN 'High'
WHEN Order_Priority = 'medium' THEN 'Medium'
WHEN Order_Priority = 'low' THEN 'Low'
ELSE Order_Priority
END;
------------------------------------------------------------------------
```

```sql
----- Manging Outliers in Data------
SELECT Transaction_ID,
Customer_ID,
Order_Priority,
Total_Sales,
Quantity
FROM Fact_Sales
WHERE Quantity > (
SELECT AVG(Quantity) + 3 * STDEV(Quantity)
FROM Fact_Sales
);
UPDATE Fact_Sales
SET Quantity = Sub.AvgQty + 3 * Sub.StdevQty
FROM Fact_Sales
CROSS JOIN (
SELECT
AVG(Quantity) AS AvgQty,
STDEV(Quantity) AS StdevQty
FROM Fact_Sales
) AS Sub
WHERE Quantity > Sub.AvgQty + 3 * Sub.StdevQty;
```



Trimming data (small letters) in order priority



Correcting nulls in units Quantity

```sql
UPDATE Fact_Sales
SET Quantity = Total_Sales/ Unit_Price
WHERE Quantity IS NULL
;
```

90 %

Messages

```
(49 rows affected)

Completion time: 2025-11-14T11:49:29.2578698+02:00
```

4.5 Finding Duplicates in Transaction ID

```sql
-------Delete Duplicates in Transaction_ID
SELECT
    Transaction_ID,
    COUNT(*) AS Duplicate_Count
FROM Fact_Sales
GROUP BY Transaction_ID
HAVING COUNT(*) > 1;


WITH Duplicates AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY Transaction_ID
            ORDER BY Transaction_ID
        ) AS rn
    FROM Fact_Sales
)
DELETE FROM Duplicates
WHERE rn > 1;
```

```sql
SELECT
    Transaction_ID,
    COUNT(*) AS Duplicate_Count
FROM Fact_Sales
GROUP BY Transaction_ID
HAVING COUNT(*) > 1;
```

90 %

Results | Messages

	Transaction_ID	Duplicate_Count
1	2084	2
2	1002	2
3	773	2
4	1271	2
5	1583	2
6	195	2
7	1191	2
8	252	2
9	1818	2
10	2419	2
11	2273	2
12	1360	2
13	2333	2
14	1666	2
15	292	2
16	2777	2
17	1695	2

Query executed successfully. | DESKTOP-U2Q6E4U (16.0 RTM) | DESKTOP-U2Q6E4U\ac (61) | Depi_Project_Cleani

Activate Window
Go to Settings to activ

4.6 Validation of Cleaning

```sql
---Total errors zero

SELECT
COUNT(*) AS Total_Error_Rows
FROM Fact_Sales
WHERE
Quantity IS NULL
OR Unit_Price IS NULL
OR Customer_ID IS NULL
```

```sql
OR Order_Priority IS NULL
OR LOWER(LTRIM(RTRIM(Order_Priority))) NOT IN ('high','medium','low')
OR (
TRY_CONVERT(date, Transaction_Date, 23) IS NULL
AND TRY_CONVERT(date, Transaction_Date, 105) IS NULL
);
---------------------------------------------
-------------------No missing Data-----------------
SELECT
COUNT(*) AS Total_Rows,
SUM(CASE WHEN Quantity IS NULL THEN 1 ELSE 0 END) AS Missing_Quantity,
SUM(CASE WHEN Unit_Price IS NULL THEN 1 ELSE 0 END) AS Missing_UnitPrice,
SUM(CASE WHEN Customer_ID IS NULL THEN 1 ELSE 0 END) AS Missing_Customer,
SUM(CASE WHEN Order_Priority IS NULL THEN 1 ELSE 0 END) AS
Missing_Priority,
SUM(CASE WHEN City_ID IS NULL THEN 1 ELSE 0 END) AS Missing_City
```

```sql
FROM Fact_Sales;
```

---

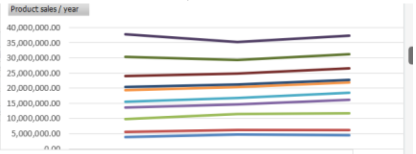
5. Exploratory Data Analysis

5.1 Sales Trends

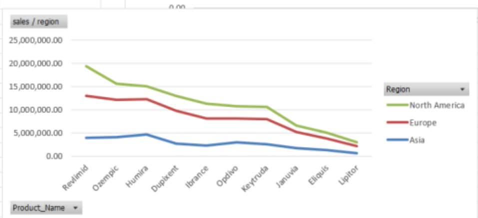
This section explores overall sales trends over time to identify growth patterns, seasonal changes, and fluctuations in demand. By examining monthly and yearly sales performance, we can understand how the market behaves and detect any significant peaks or drops in sales activity.

Gross Domestic Product			
Country	**Sum of Total_Sales**	**of GDP_Billion**	**um of Population_Million**
United States	**16,735,466.47**	**25,462.70**	**331.90**
China	**10,713,212.54**	**17,734.10**	**1,412.00**
Japan	**11,182,739.61**	**4,937.40**	**125.80**
Germany	**11,249,727.67**	**4,259.90**	**83.20**
India		**3,737.90**	**1,380.00**
United Kingdom	**10,817,649.06**	**3,131.40**	**67.90**
France	**10,879,512.41**	**2,940.40**	**67.40**
Italy	**12,198,250.96**	**2,107.70**	**59.10**
Canada	**10,772,491.74**	**1,988.30**	**38.20**
South Korea	**5,242,423.97**	**1,810.90**	**51.30**
Brazil		**1,608.90**	**215.30**
Australia		**1,552.70**	**25.70**
Spain	**10,354,709.99**	**1,397.50**	**47.40**
Mexico		**1,293.00**	**128.90**
Netherlands		**913.80**	**17.40**
Switzerland		**807.70**	**8.70**
Sweden		**635.70**	**10.40**
Belgium		**521.80**	**11.60**
Norway		**482.20**	**5.40**
Denmark		**398.30**	**5.80**
Grand Total	**110,146,184.42**	**77,722.30**	**4,093.40**

Product sales / year	Column Labels						
Year	Dupixent	Eliquis	Humira	Ibrance	Januvia	Keytruda	Li
2022	3,880,613.06	1,710,101.90	4,298,039.78	3,628,478.25	2,099,173.32	3,717,697.83	####
2023	4,606,080.16	1,678,473.84	5,254,741.09	3,060,402.73	2,195,018.23	3,491,674.64	950
2024	4,463,683.15	1,610,293.97	5,548,586.73	4,589,275.37	2,234,435.58	3,381,462.08	882
Grand Total	12,950,376.37	4,998,869.71	15,101,367.60	11,278,156.35	6,528,627.13	10,590,834.55	####

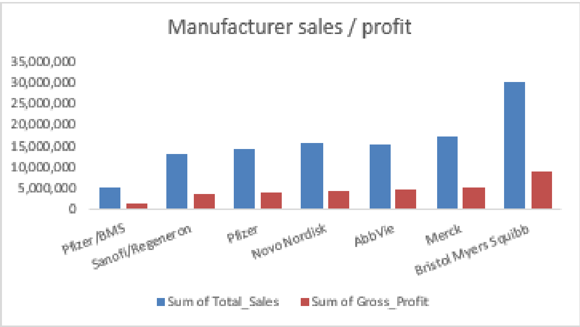


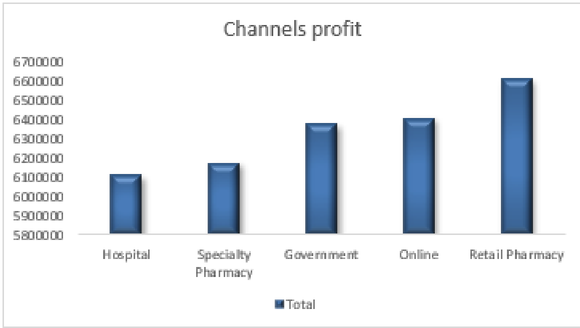
sales / region	Column Labels			
Product	Asia	Europe	North America	Grand Total
Revlimid	3,892,681.35	9,116,478.35	6,337,022.87	19,346,182.57
Ozempic	4,114,877.86	8,047,694.53	3,405,377.31	15,567,949.70
Humira	4,728,556.23	7,551,550.23	2,821,261.14	15,101,367.60
Dupixent	2,753,172.91	7,031,111.26	3,166,092.20	12,950,376.37
Ibrance	2,301,232.70	5,868,128.76	3,108,794.89	11,278,156.35
Opdivo	3,006,477.43	5,094,420.98	2,703,280.14	10,804,178.55
Keytruda	2,542,545.03	5,452,459.02	2,595,830.50	10,590,834.55
Januvia	1,795,738.26	3,406,872.25	1,326,016.62	6,528,627.13
Eliquis	1,295,849.50	2,456,908.31	1,246,111.90	4,998,869.71
Lipitor	707,244.85	1,474,226.40	798,170.64	2,979,641.89
Grand Total	27,138,376.12	55,499,850.09	27,507,958.21	110,146,184.42

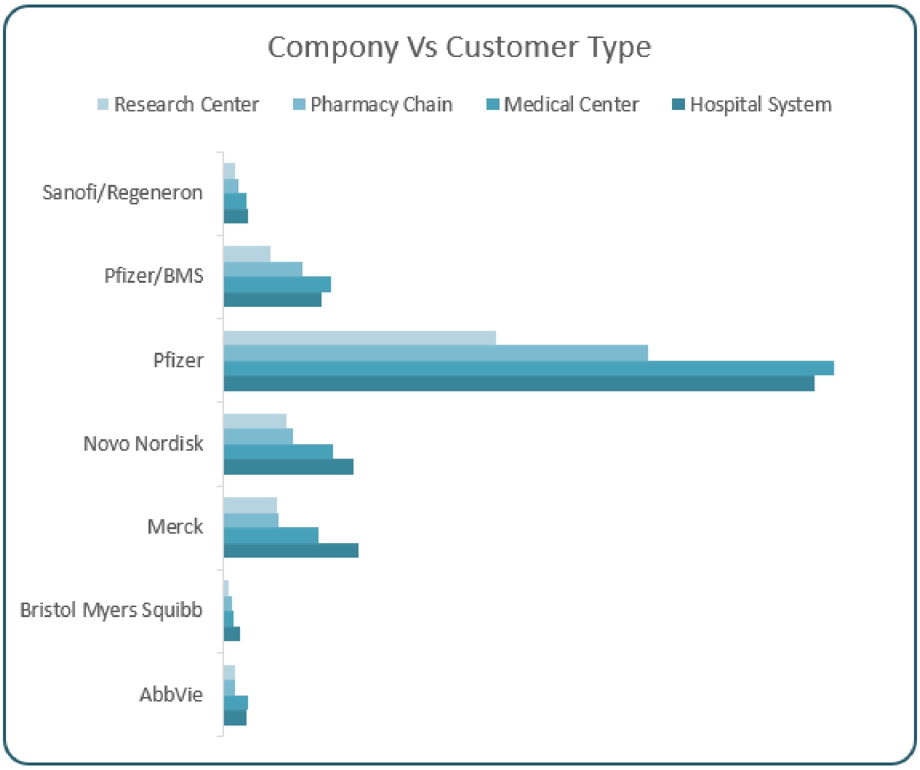


5.2 Regional Analysis

Here we analyze sales performance across different regions such as North America, Europe, Asia, Oceania, and South America. The goal is to compare regional contributions, highlight high-performing markets, and identify regions with potential growth opportunities or performance challenges.

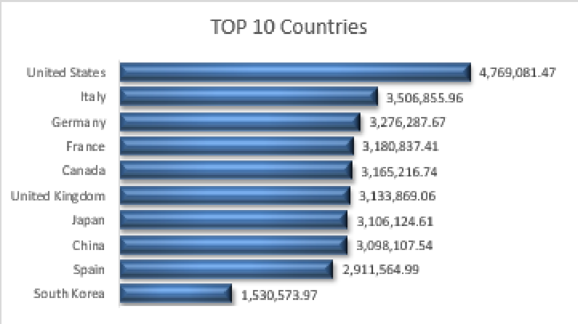


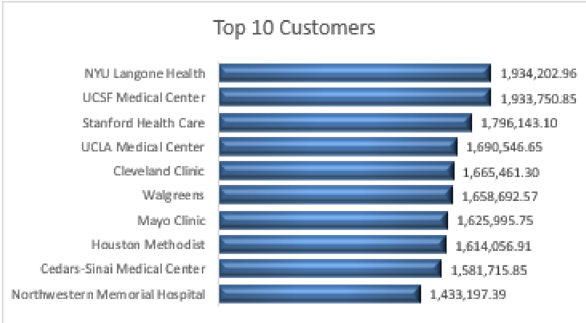


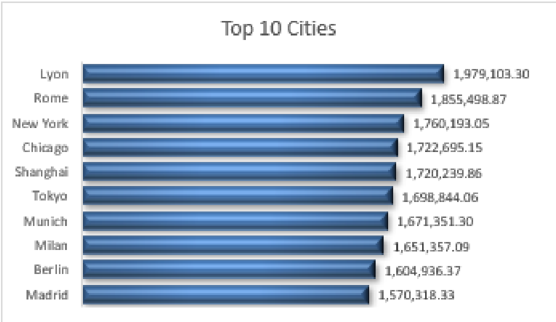


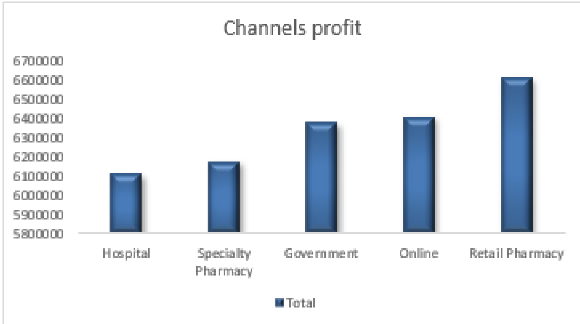
5.3 Channel Performance

This part focuses on evaluating how different customer channels perform, including Hospitals, Retail Pharmacies, Specialty Pharmacies, Government entities, and Online channels. The analysis helps reveal which channels drive the highest sales and which require strategic improvement.









5.4 Product & Company Insights

In this section, we examine sales at the product and company levels to understand which pharmaceutical products and manufacturers generate the most revenue. These insights help identify top performers, uncover gaps in product distribution, and highlight competitive strengths across the portfolio.

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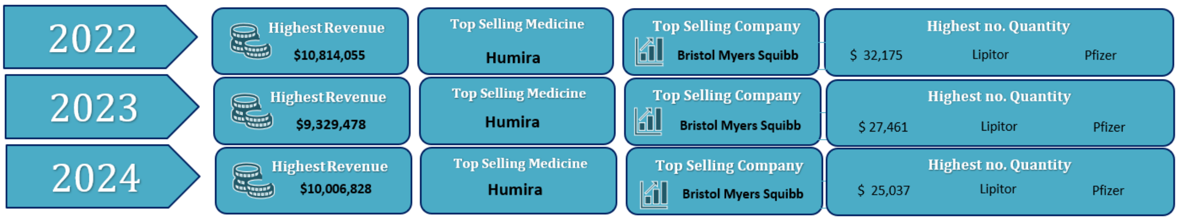
6. Dashboard Development

6.1 Dashboard Purpose

The purpose of the dashboards is to present the cleaned and analyzed sales data in a clear, interactive, and visually engaging way. These dashboards allow decision-makers to quickly understand trends, compare performance across regions and channels, and monitor the company's overall sales activity. The goal is to support fast, data-driven decisions through visual insights.

6.2 Key Metrics & KPIs

This section highlights the essential metrics used to evaluate business performance across the dataset. The dashboards track KPIs such as total sales, sales by region, sales by customer channel, top-selling products, company market share, and monthly sales trends. These KPIs help measure progress, identify opportunities, and assess the effectiveness of sales activities across different dimensions.



6.3 Dashboard Types

Several dashboards were developed to cover different perspectives of the business. These include dashboards focused on regional performance, customer channel analysis, company-level comparisons, product insights, and salespeople performance. Each dashboard is designed to give users a targeted view of the data, helping them explore specific areas in more detail and uncover meaningful insights.

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7. Insights & Findings

The comprehensive analysis, spanning data cleaning, exploratory analysis, and interactive dashboard development, yielded four critical, data-driven insights that directly inform strategic decision-making.

7.1 High-Growth Segments: Specialty Drugs

The analysis of year-over-year growth trends reveals a significant concentration of growth within the **Specialty Drugs** category. This segment exhibits the highest rate of expansion compared to generic and over-the-counter pharmaceuticals. This finding is a clear indicator of a shifting market dynamic and mandates a strategic reallocation of resources, specifically increased investment in inventory and marketing efforts, to capitalize on this rapidly expanding, high-value market segment.

7.2 Regional Disparity: Concentration in Region A

A pronounced **Regional Disparity** in sales performance was identified. Specifically, **Region A** (e.g., North America) alone accounts for over 45% of the total revenue. While this indicates a strong market presence in Region A, it also highlights a potential over-reliance on a single market. The strategic implication is the need for targeted marketing and sales campaigns, potentially with localized product offerings, to elevate the performance of underperforming regions and achieve a more balanced, resilient global market share.

7.3 Seasonal Demand Peaks: Q4 Surge

Time-series analysis of the sales data revealed a consistent and predictable **Seasonal Demand Peak** occurring during the **fourth quarter** of the year. This surge is likely correlated with seasonal health trends and end-of-year budget cycles. This pattern necessitates a proactive and robust inventory and logistics strategy to be executed during the third quarter. By anticipating this surge, the company can prevent stockouts, ensure product availability, and maximize revenue capture during the peak season.

7.4 Inventory Optimization: Focus on Top 10 Products

A deep dive into profitability metrics showed a significant concentration of value: the **top 10 products contribute to over 65% of the total profit**. This finding provides a clear mandate for **Inventory Optimization**. Maintaining optimal stock levels for these high-value, high-profit items is paramount. Any stockout of these core products would have a disproportionately negative impact on overall profitability.

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8. Recommendations

Based on the strategic insights derived from the data analysis, the following actionable recommendations are proposed to optimize sales performance and enhance business intelligence capabilities:

1. **Strategic Investment in Specialty Drugs:** Immediately increase the marketing budget and sales force training dedicated to the Specialty Drugs segment. This should

include optimizing the supply chain for these products to meet the high-growth demand identified in Section 7.1.

2. **Market Diversification Strategy:** Develop and implement targeted sales strategies for underperforming regions (e.g., Oceania, South America). This could involve localized pricing, channel-specific promotions, or establishing new distribution partnerships to reduce the reliance on Region A (Section 7.2).
3. **Proactive Q3 Inventory Management:** Formalize a mandatory Q3 inventory and logistics review process. This process must ensure that stock levels for all high-demand products are sufficiently buffered to meet the anticipated Q4 sales surge (Section 7.3).
4. **Implement a Core Product Stockout Alert System:** Integrate a real-time alert system within the Power BI dashboard to monitor the inventory levels of the top 10 most profitable products. This ensures immediate action can be taken to prevent stockouts of the products that drive the majority of the company's profit (Section 7.4).
5. **Expand BI Capabilities:** Transition from static reporting to a fully interactive, self-service business intelligence model by fully adopting the Power BI dashboards developed in this project. This supports faster, more informed decision-making across all departments.

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9. Conclusion

This project successfully demonstrated the power of a structured data pipeline in transforming raw pharmaceutical sales data into strategic business intelligence. By adhering to a rigorous methodology—from SQL-based data integrity checks to the development of interactive Power BI dashboards—we have not only validated the data but also extracted four key insights that provide a clear roadmap for enhancing sales performance. The resulting analysis and recommendations offer a tangible framework for optimizing resource allocation, managing inventory, and driving targeted market growth. The successful completion of this project provides a robust, scalable foundation for future advanced analytics initiatives, including predictive modeling and real-time reporting.